

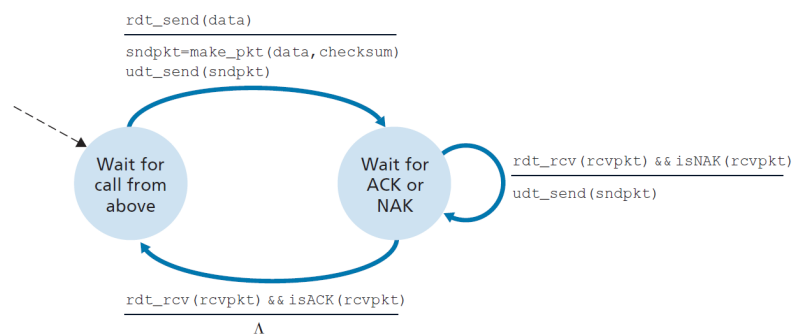
Reliable Data Transfer Protocols (RDT)

1 Reliable Data Transfer over a Channel with Bit Errors: rdt2.0

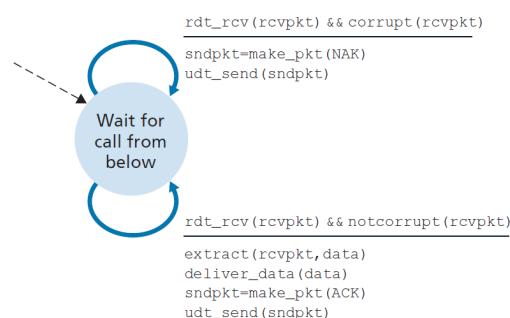
A realistic network channel may introduce **bit errors** during packet transmission. The rdt2.0 protocol handles this using **ARQ (Automatic Repeat reQuest)** mechanisms.

Key Features of rdt2.0

- **Error Detection:** Using checksums to detect corrupted packets.
- **Receiver Feedback:** Sends **ACK** if packet is correct or **NAK** if corrupted.
- **Retransmission:** Sender retransmits corrupted packets.
- **Stop-and-Wait:** Sender waits for receiver feedback before sending the next packet.



a. rdt2.0: sending side



b. rdt2.0: receiving side

Figure 1: rdt2.0 FSM overview. Sender has two states (waiting for data and waiting for ACK/NAK). Receiver sends ACK/NAK based on packet correctness.

1.1 FSM Event Handling

1. Sender:

- If ACK received → proceed to next data packet.
- If NAK received → retransmit last packet.

2. Receiver:

- Sends ACK for correctly received packet.
- Sends NAK for corrupted packet.

1.2 Limitations of rdt2.0

Problem & Solution

- **Problem:** ACK/NAK packets can be corrupted.
- **Solution:** Introduce **sequence numbers** to detect duplicate or missing packets.

2 RDT with Sequence Numbers: rdt2.1 and rdt2.2

Key Idea

Use a 1-bit **sequence number** to distinguish between original and retransmitted packets.

2.1 rdt2.1

- Sender and receiver FSMs now have **two states each** (sequence 0 or 1).
- Receiver sends ACK for correctly received packets, NAK for corrupted packets.

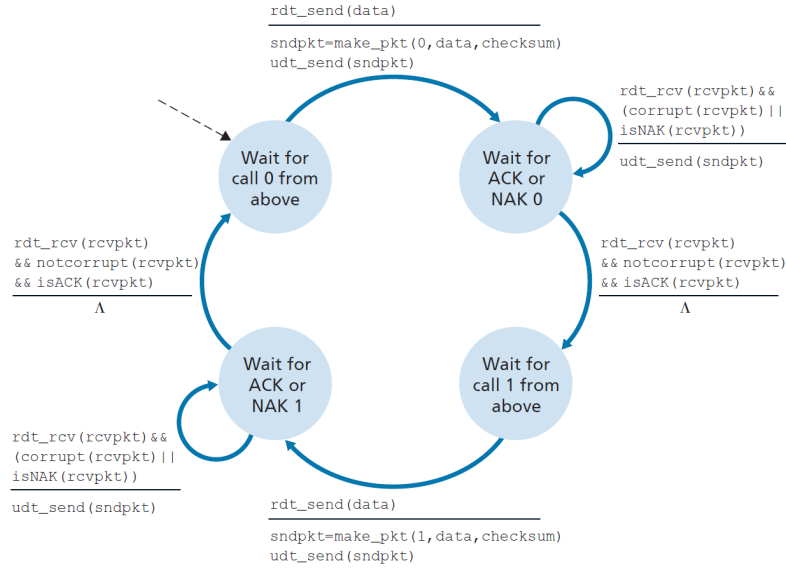


Figure 2: rdt2.1 Sender FSM: alternating between sending sequence 0 and 1.

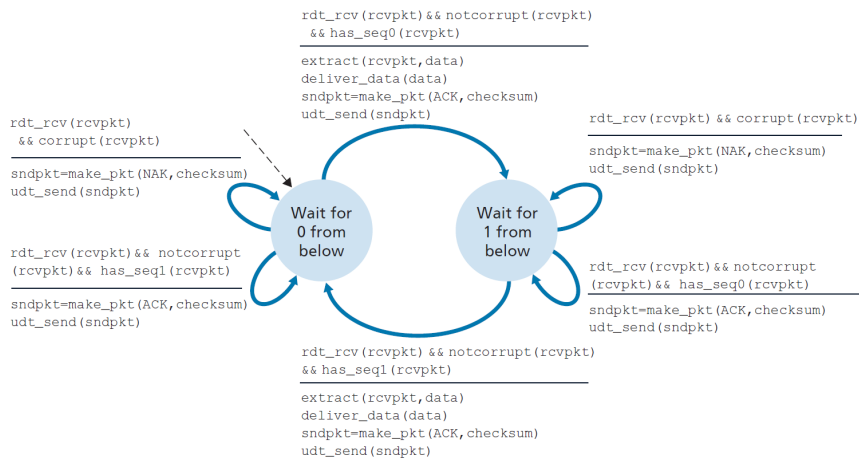


Figure 3: rdt2.1 Receiver FSM: checks sequence number to detect retransmissions.

2.2 rdt2.2 (NAK-Free)

- Instead of NAK, receiver sends **ACK for last correctly received packet**.
- Sender uses sequence number to detect duplicate or missing packets.
- Simplifies protocol design and reduces control packet traffic.

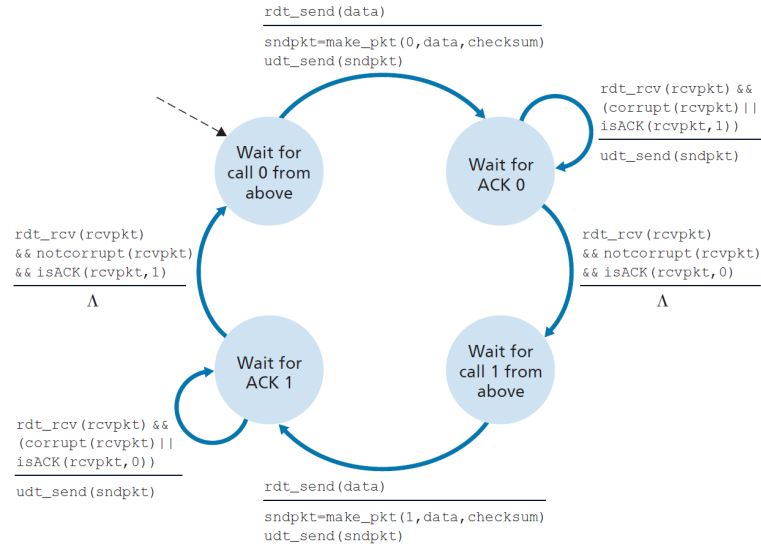


Figure 4: rdt2.2 Sender FSM: uses sequence numbers to handle retransmissions.

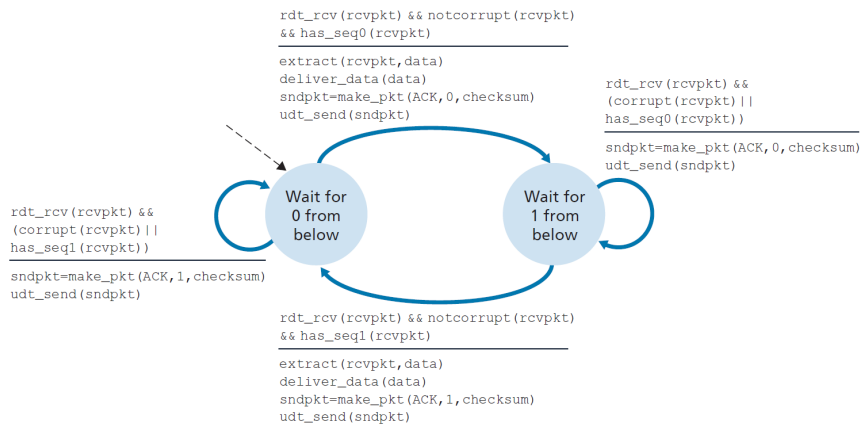


Figure 5: rdt2.2 Receiver FSM: NAK-free; sends ACK for last correctly received packet.

3 Reliable Data Transfer over Lossy Channels: rdt3.0

Channels may **lose packets** in addition to bit corruption. rdt3.0 addresses this using **timers** and retransmissions.

New Challenges in rdt3.0

- Detect packet loss if ACK not received within a timeout period.
- Retransmit data packet after timeout.
- Sequence numbers handle duplicates from retransmissions.
- Known as the **alternating-bit protocol**.

3.1 Stop-and-Wait with Timers

1. Start timer when sending a packet.
2. If ACK not received before timeout $\rightarrow$ retransmit packet.
3. Sequence numbers prevent duplicate data delivery.

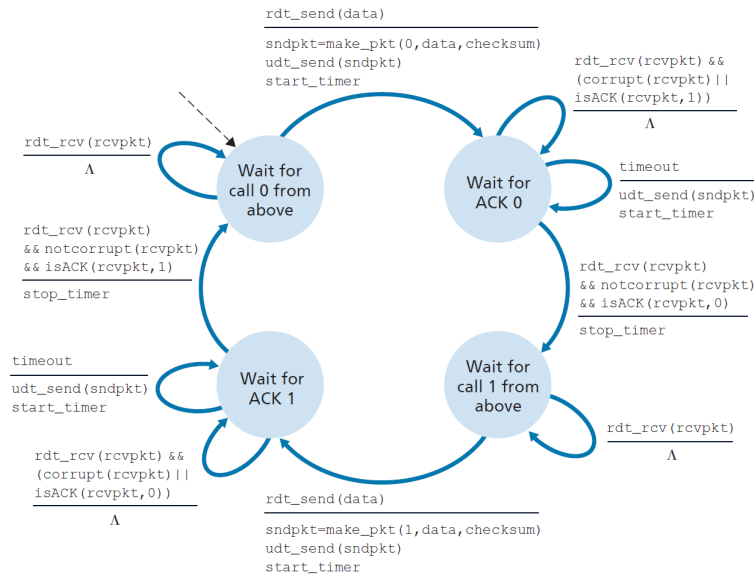


Figure 6: rdt3.0 Sender FSM: includes timer mechanism for lost packet recovery.

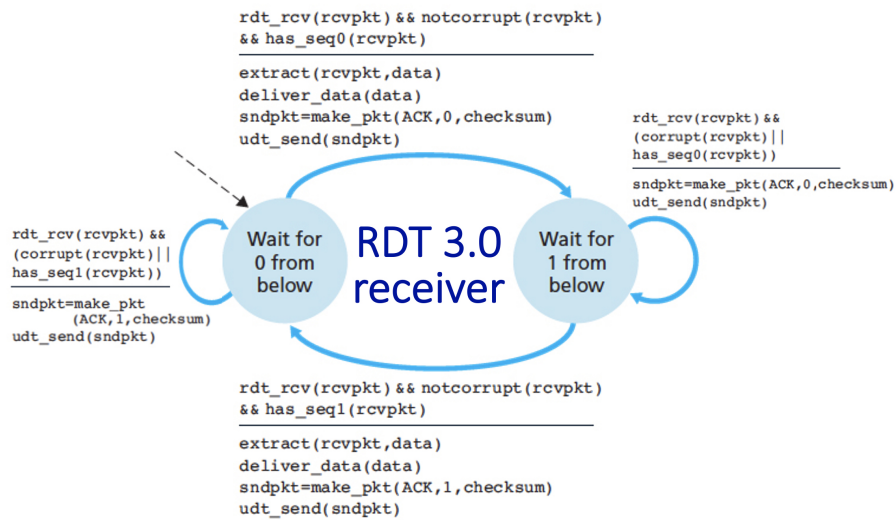


Figure 7: rdt3.0 Receiver FSM: handles duplicate packets and lost ACKs using sequence numbers.